YUWON KIM

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HCI · VIRTUAL REALITY · MULTIMODAL ANALYTICS

A Human-Computer Interaction (HCI) researcher who transforms multimodal data, sensed through virtual reality (VR) and wearable devices, into actionable insights that support human decision-making. My research includes translating behavioral data in VR into indicators of cognitive deficits, as well as analyzing stress during interactions with virtual avatars. As Principal Investigator or Lead Researcher, I built and validated these virtual environment systems through multidisciplinary collaboration with clinical experts at Asan Medical Center (Seoul), engineers at the Korea Institute of Industrial Technology (KITECH), and international academic partners at the Zurich University of Applied Sciences.

Education

PhD Candidate, Department of Applied Artificial Intelligence — 2022.03.01. ~ 2027.02 (expected)
Seoul National University of Science and Technology, Seoul, Korea — GPA: 4.18 / 4.5

- Advisor: Prof. Kyoungwon Seo
- Thesis: (in progress) An Information Fusion Framework of VR-Based Digital Biomarkers for Early Screening of Mild Cognitive Impairment

BS, Optometry (Primary Major) & ICT Artificial Intelligence (Double Major) — 2016.03.02. ~ 2022.02.17.
Seoul National University of Science and Technology, Seoul, Korea — GPA: 3.5 / 4.5

Research Projects

LLM-Based Clinical Decision Support Tool for Alzheimer's Disease via Multimodal Reasoning Integrating VR, MRI, and Neuropsychological Tests — 2025.09 ~ Present
Principal Investigator, National Research Foundation of Korea (25,000k KRW)

- Independently secured competitive national research funding as sole Principal Investigator during doctoral study, to design an LLM-based explainable AI (XAI) framework that interprets VR digital biomarkers and generates explanations for clinicians to support clinical decision-making.
- Extending the VR-based digital biomarker framework into a multimodal reasoning clinical decision support tool integrating multimodal data, such as MRI and neuropsychological test data.

On-Device AI-Based Self-Guided Hemorrhage Management System for Vascular Intervention using Vascular Image Data — 2025.09 ~ 2026.03
Lead Researcher; Ministry of Trade, Industry and Energy (3,600m KRW)

- Designed the methodology and led the implementation of an AI-embedded robotic system that automatically assesses hemorrhage status and performs autonomous hemorrhage control.
- Mentored the research team that produced work published at HCI Korea 2026, with a patent application in preparation on ultrasound-based vascular puncture site detection and AI-driven action generation for clinical decision support.

Development of AI-Based Analysis and Learning Model for Social Interaction Abilities Using Immersive Content for Individuals with Developmental Disabilities — 2025.04 ~ 2025.12
Lead Researcher; Ministry of SMEs and Startups (50m KRW)

- Designed and evaluated a virtual-agent-based task to assess social interaction abilities in individuals with developmental disabilities, based on their interaction styles with the agents.
- Mentored the research team that produced work published at ACM CHI 2025 and HCI Korea 2025.

VR, EP, EEG, MRI Data Aggregation for Digital Dementia Biomarker 2021.09 ~ 2024.02
Lead Researcher; Ministry of Science and ICT (1,200m KRW)

- Led the multimodal data-aggregation pipeline integrating VR-based behavioral markers, evoked potentials (EP), electroencephalogram (EEG), and magnetic resonance imaging (MRI) into a unified digital biomarker framework for early screening of mild cognitive impairment.
- First-authored the resulting research, with work published in Journal of Medical Internet Research (SCIE, Q1) and Sensors (SCIE, Q1).

Developing Digital Biomarker for Early Diagnosis of Dementia in VR 2021.09 ~ 2024.02
Lead Researcher; Ministry of Science and ICT (550m KRW)

- Led development and validation of a VR-based virtual kiosk cognitive test and eye-movement/hand-movement tracking system for early dementia screening, from graduate school entry through project completion.
- First-authored the resulting research, with work published in Information Processing & Management (SSCI, Q1) and Journal of Medical Internet Research (SCIE, Q1).

Journal Articles

Click any entry below to view the full paper.
† = Equal Contribution;

[J4] Park, B.† , **Kim, Y.**† , Park, J., Choi, H., Kim, S. E., Ryu, H., & Seo, K. (2024). Integrating biomarkers from virtual reality and magnetic resonance imaging for the early detection of mild cognitive impairment using a multimodal learning approach: validation study. *Journal of Medical Internet Research, 26*, e54538. **[SCIE, Q1, Top 3%]**

[J3] Kim, D†., **Kim, Y.**†, Park, J., Choi, H., Ryu, H., Loeser, M., & Seo, K. (2024). Exploring the relationship between behavioral and neurological impairments due to mild cognitive impairment: Correlation study between virtual kiosk test and EEG-SSVEP. *Sensors, 24*(11), 3543. **[SCIE, Q1]**

[J2] Kim, S. Y., Park, J., Choi, H., Loeser, M., Ryu, H., & Seo, K. (2023). Digital marker for early screening of mild cognitive impairment through hand and eye movement analysis in virtual reality using machine learning: First validation study. *Journal of Medical Internet Research, 25*, e48093. **[SCIE, Q1, Top 3%]**

[J1] Kim, S. Y., Park, H., Kim, H., Kim, J., & Seo, K. (2022). Technostress causes cognitive overload in high-stress people: Eye tracking analysis in a virtual kiosk test. *Information Processing & Management*, 59(6), 103093. **[SSCI, Q1, Top 5%]**

Conference Papers

Click any entry below to view the full paper

[C10] Kang, J., **Kim, Y.**, Won, C., & Seo, K. (2026). Sparse–Dense Integrated Auto-Prompt SAM for Accurate Carotid Artery Segmentation in Ultrasound Imaging. In *Proceedings of the 2026 HCI Korea Conference*.

[C9] Kim, Y. (2025, April). Beyond Missing Data: A Multimodal Approach Using VR-EEG-MRI (VEEM) Biomarkers for Detecting MCI. In *Extended Abstracts of the ACM CHI Conference on Human Factors in Computing Systems* (Student Research Competition). **[Research Competition Second-Place Award; acceptance rate: 15%]**

[C8] You, D., **Kim, Y**., & Seo, K. (2025, April). Tailored Virtual Agent Guidance for Stress Management: Comparing Directive and Non-Directive Approaches by Alexithymia Status. In *Extended Abstracts of the ACM CHI Conference on Human Factors in Computing Systems*. **[acceptance rate: 32.8%]**

[C7] Cha, S. E., **Kim, Y**., & Seo, K. (2025). Effects of Tool Description Language on Language Model's Tool Calling in Korean Dialogue. In *Proceedings of the 2025 HCI Korea Conference*.

[C6] You, D., **Kim, Y**., & Seo, K. (2025). Digital Mental Health Intervention with Virtual Agent: Impact of Guidance Approaches on Stress Reduction in Young Adults. In *Proceedings of the 2025 HCI Korea Conference*.

[C5] Keum, H., **Kim, Y**., You, D., & Seo, K. (2025). A Deep Learning Model for Stress Level Recognition Utilizing Neutral and Stress State of Remote Photoplethysmography. In *Proceedings of the 2025 HCI Korea Conference*.

[C4] Kim, Y., Park, J., Choi, H., Loeser, M., Ryu, H., & Seo, K. (2024, May). Decoding Behavior: Utilizing Virtual Reality Digital Marker and Machine Learning for Early Detection of Mild Cognitive Impairment. In *Extended Abstracts of the ACM CHI Conference on Human Factors in Computing Systems*. **[acceptance rate: 33.8%]**

[C3] Park, B., **Kim, Y**., Park, J., Choi, H., Kim, S. E., Ryu, H., & Seo, K. (2024, May). Exploring the Multimodal Integration of VR and MRI Biomarkers for Enhanced Early Detection of Mild Cognitive Impairment. In *Extended Abstracts of the ACM CHI Conference on Human Factors in Computing Systems*. **[acceptance rate: 33.8%]**

[C2] Kim, S. Y., Park, B., Kim, D., Choi, H., Park, J., Ryu, H., & Seo, K. (2024). Early Screening of Mild Cognitive Impairment Using Multimodal VR-EP-EEG-MRI (VEEM) Biomarkers via Machine Learning. In *2024 International Conference on Electronics, Information, and Communication (ICEIC)*.

[C1] Kim, S. Y., Park, J., Choi, H., Ryu, H., & Seo, K. (2023). Virtual Kiosk Test for Early Screening of Mild Cognitive Impairment through Eye Tracking Data Analysis. In *Proceedings of the 2023 HCI Korea Conference*.

Patents

[P3] Seo, K., **Kim, Y.** (2026). Understanding Vascular Puncture Sites in Ultrasound Images and Action-generation Artificial Intelligence for Clinical Decision Support System. (in progress)

[P2] Seo, K., **Kim, S. Y.** (2025). VR(Virtual Reality) Digital Biomarker based Clinical Decision Support System and Method using Artificial Intelligence for Early Diagnosis of Alzheimer's Disease. *KR10-2025-0022344*. (published 2025.02.17)

[P1] Seo, K., **Kim, S. Y.** (2024). Eye tracking data calibration method and eye tracking data calibration system. *KR10-2024-0048131*. (published 2024.04.15)

Research Grants & Awards

NRF, Research Subsidies for Ph.D. Candidates (PI) 25,000k KRW 2025.09.01. ~ 2026.08.31.

- Title: LLM-Based Clinical Decision Support Tool for Alzheimer's Disease via Multimodal Reasoning Integrating VR, MRI, and Neuropsychological Tests

Seoul Scholarship Foundation, Seoul AI Tech Graduate Scholarship (PI) 10,000k KRW Fall 2025

- Title: Construction of a VR-MRI-Cognitive Function Knowledge Graph and Training of a Multimodal Reasoning LLM Backbone Model as the Core Foundation for Dementia Early-Diagnosis Assistive Technology

ACM CHI 2025, Research Competition Second-Place Award 2025.04.26. ~ 2025.05.01.

- Title: Beyond Missing Data: A Multimodal Approach Using VR-EEG-MRI (VEEM) Biomarkers for Detecting MCI

Teaching

Deep Learning (Assistant) Fall 2022

- Taught deep neural network principles, mathematical foundations, and key models including autoencoders and GANs
- Guided students through a term project applying deep learning models to address real-world problems

Professional Service

Workshop Organization

- Co-Facilitator, *Augmented Educators and AI: Shaping the Future of Human-AI Collaboration in Learning*, ACM CHI 2025
- Co-Facilitator, *Medical HCI*, HCI Korea 2024
- Co-Facilitator, *Medical HCI*, HCI Korea 2023

Peer Review

- Reviewer, ACM CHI Conference on Human Factors in Computing Systems (CHI 2026)
- Reviewer, ACM Transactions on Computing for Healthcare (HEALTH)

Technical Skills

- **Programming Languages & Tools:** Python, C#
- **Machine Learning / Deep Learning:** PyTorch, TensorFlow, OpenCV, scikit-learn, multimodal learning, deep neural network design (CNNs, Segment Anything Model), model training & evaluation
- **LLM & Generative AI Application Development:** Prompt Engineering, LLM Tool/Function Calling, LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Vector Databases (FAISS, Pinecone), OpenAI API, Anthropic Claude API, Hugging Face Transformers, Multimodal LLM Reasoning
- **Conversational AI & Dialogue Systems:** Virtual/Conversational Agent Design, Dialogue Management, Human-AI Interaction Evaluation
- **VR/AR/Spatial Computing Development:** Unity, Apple Vision Pro (visionOS)
- **Data Processing & Analysis:** Image Data Labeling & Annotation (Supervisely), Eye-Tracking Analysis, EEG (SSVEP) Analysis & Preprocessing, MRI Data Processing, Remote Photoplethysmography (rPPG)
- **Knowledge Representation:** Knowledge Graph Construction (Neo4j, OWL)
- **Research & Data Tools:** Multimodal Data Fusion, Statistical Analysis (R, SPSS), Data Visualization

References

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